

DDR基础课程

第二节 - DRAM的特点

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目录

关注的问题

- DRAM相比于SRAM的关键区别是什么？
- DRAM的特点带来的设计挑战有哪些？
- 初步了解DRAM Jedec(Joint Electron Tube Engineering Council) Spec

DRAM相比于SRAM的关键区别是什么？

DRAM在片外

- 不同于SRAM，DRAM对于SoC而言是片外的，需要通过板级走线或其他方式与SoC连接

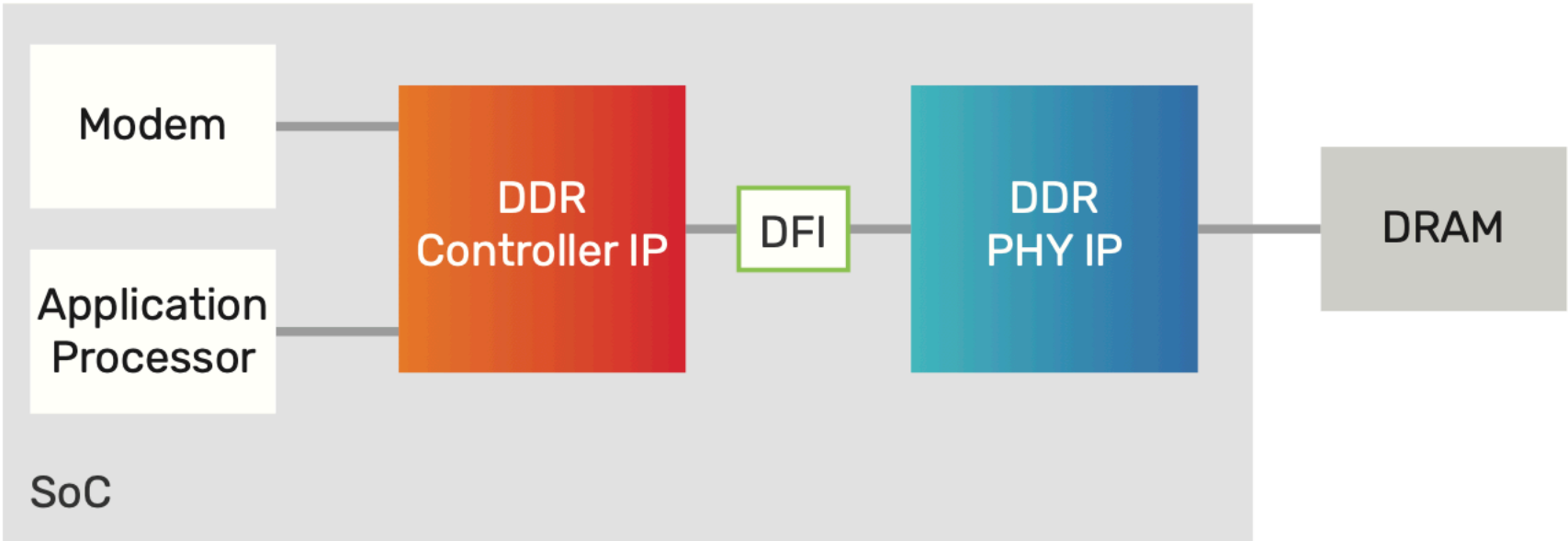
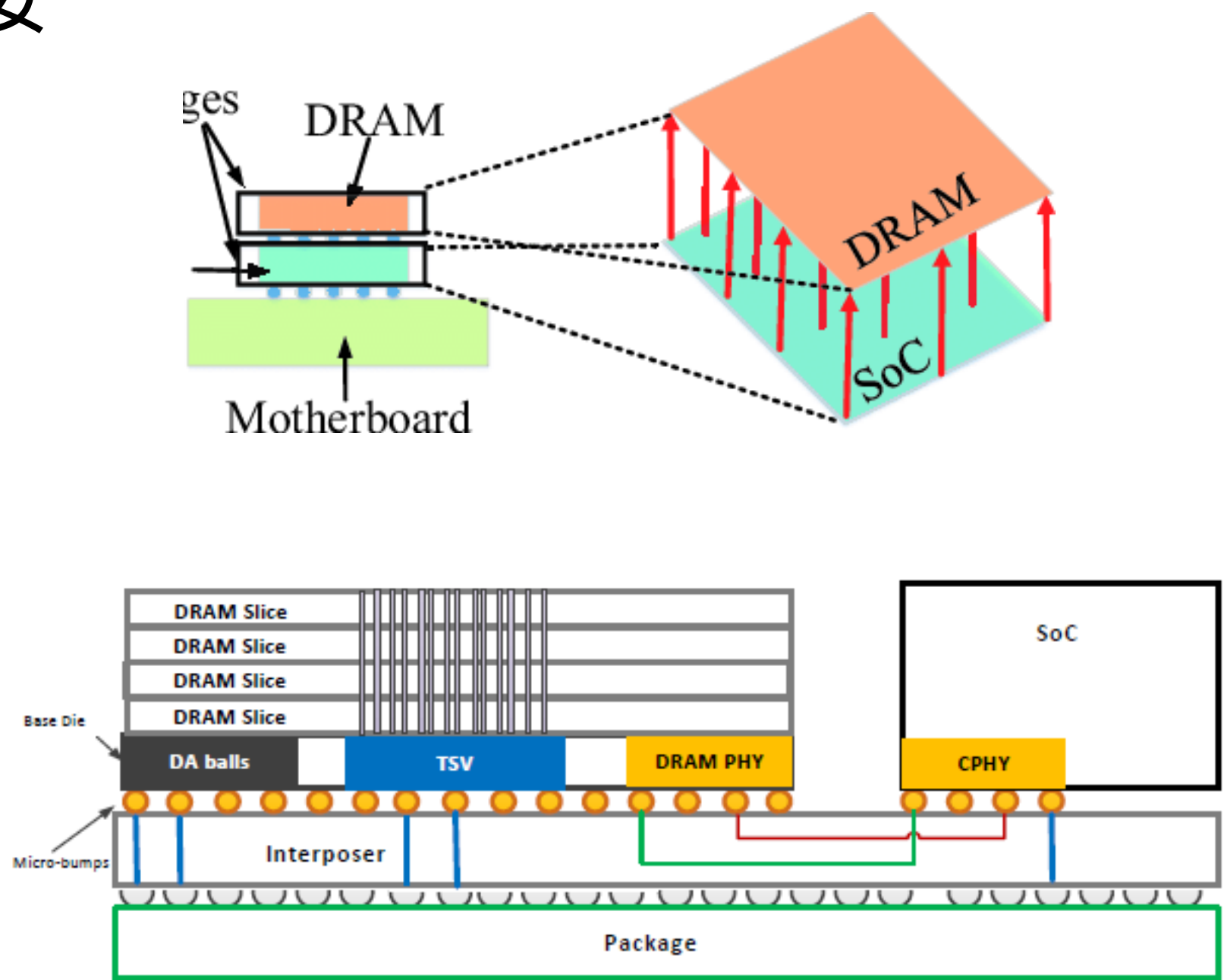
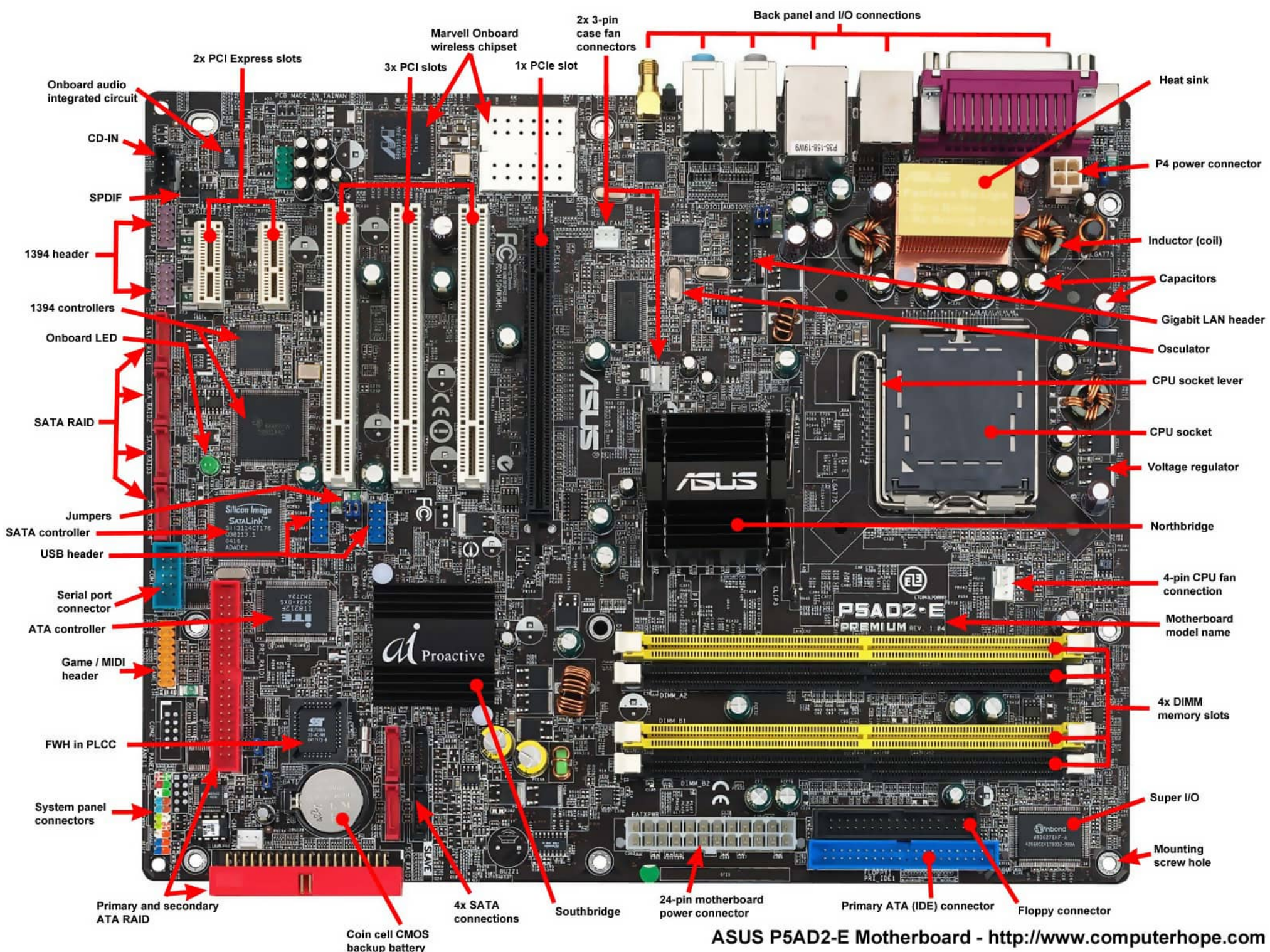


Figure 1: Example system-level block diagram



DRAM相比于SRAM的关键区别是什么？

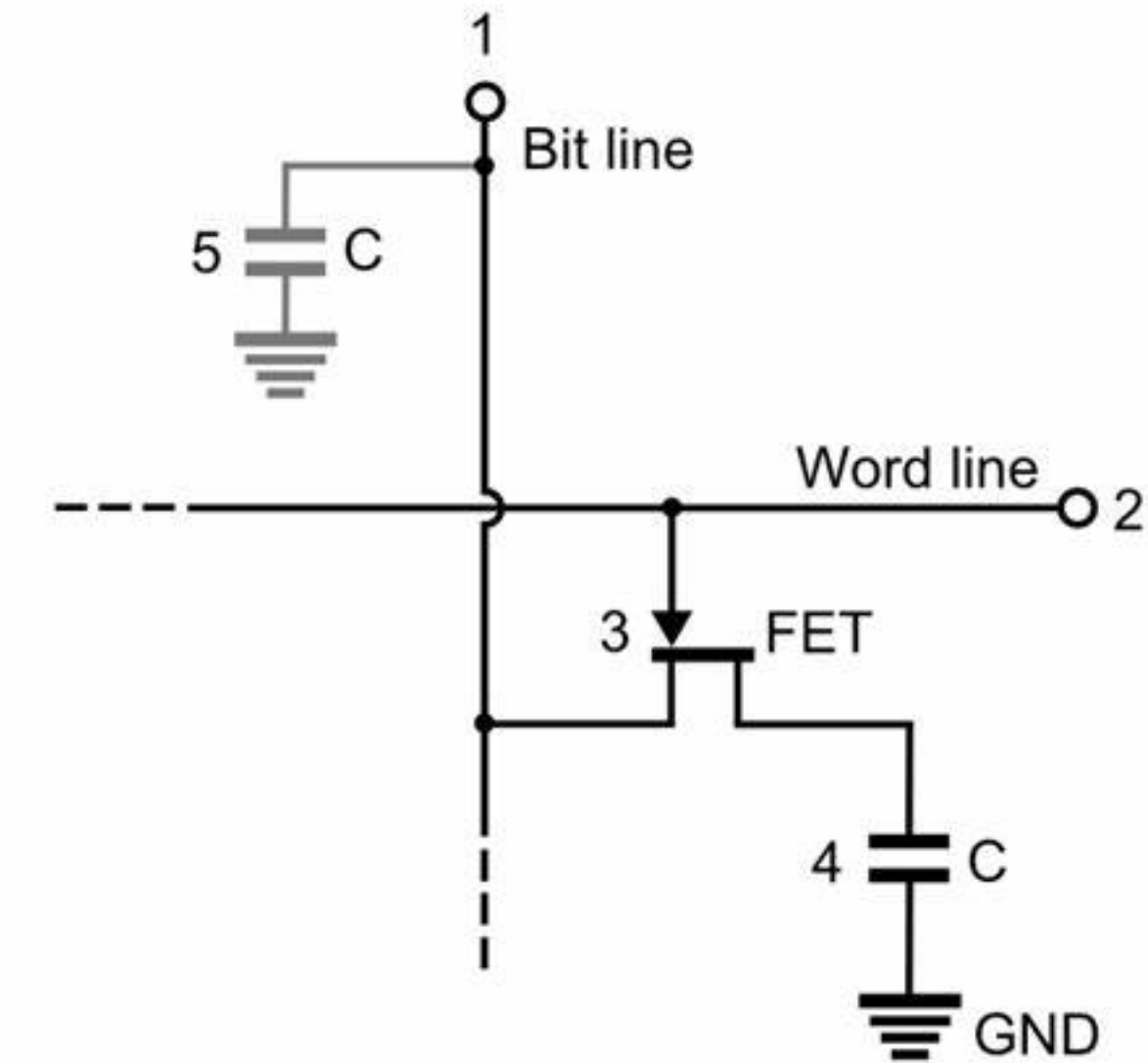
DRAM存储介质速度慢

- DRAM存储介质速度慢，且更新缓慢
- 原因来自于DRAM的存储结构
- 同样带宽要求下，需要DRAM内部并行访问多个存储介质单元
- 单个DDR命令对应的数据量较大（LP5: 32Byte）

DRAM相比于SRAM的关键区别是什么？

DRAM是动态存储器

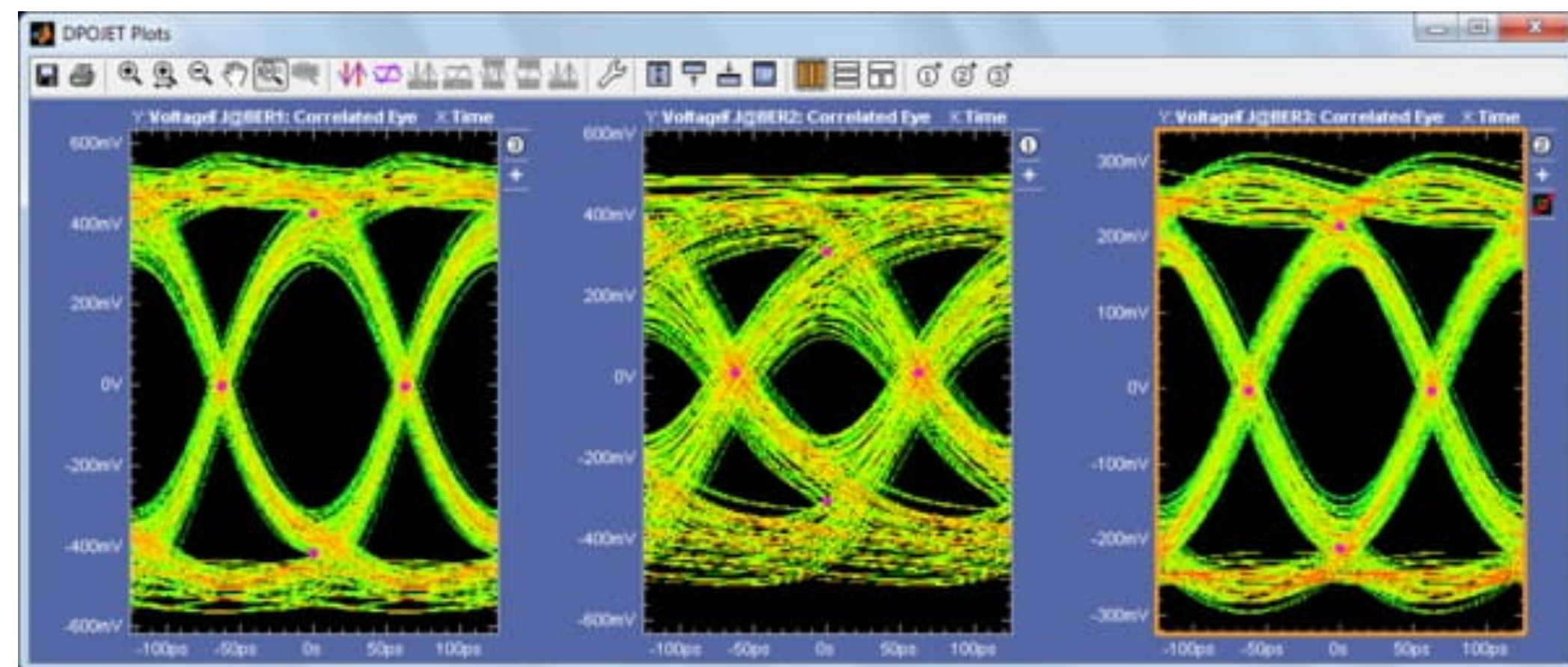
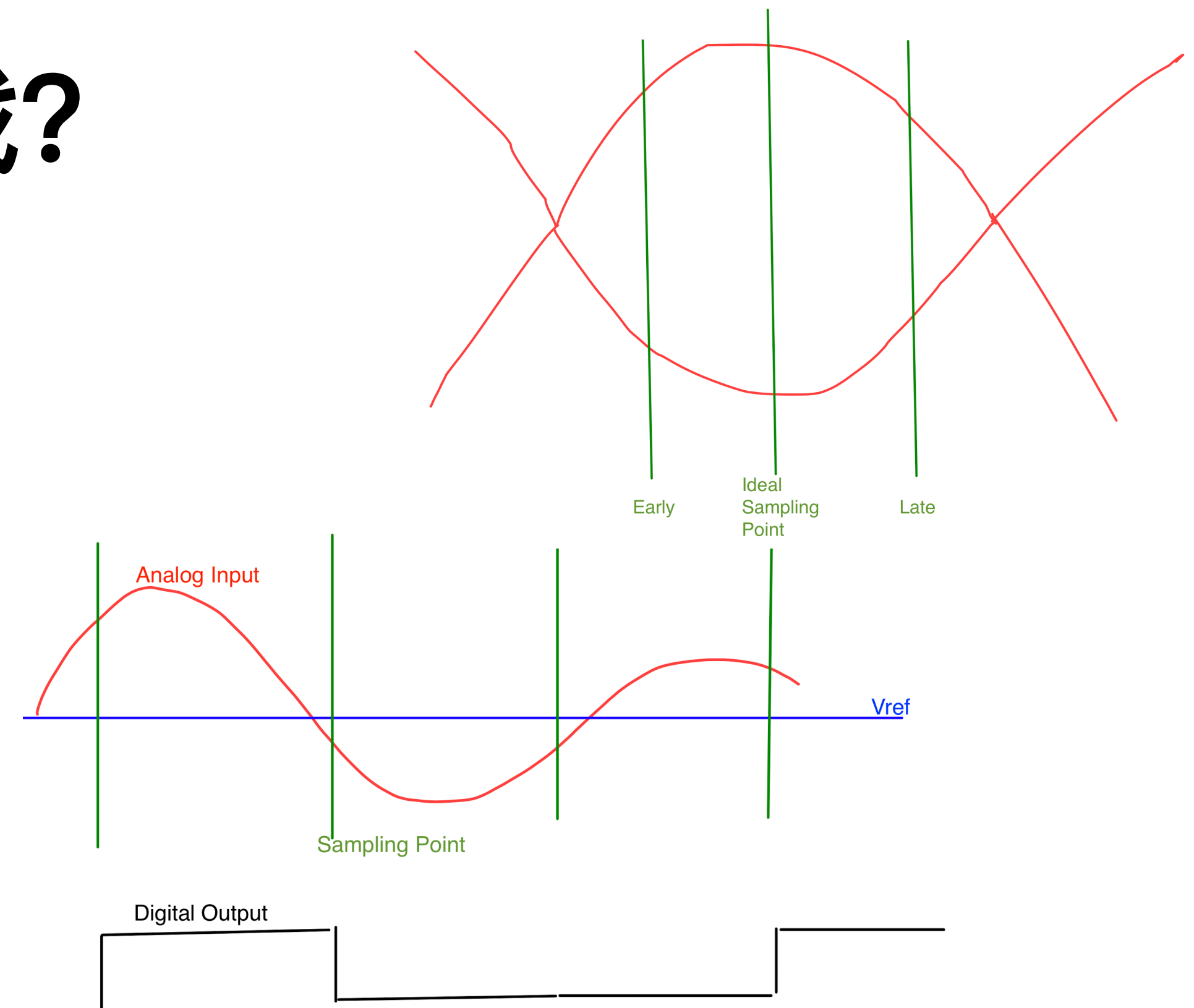
- 不同于S(Static)RAM, D(Dynamic)RAM是动态的
- 除非有其他干涉, 否则存储内容会因为时间的推移而逐渐消失
- 原因是Transistor不是理想的, 存在漏电
- 这个概念不同于“易失性(Volatile)”, 易失性指的是在没有电源的情况下其存储内容会失去



DRAM的特点带来的设计挑战?

DRAM在片外

- 相比于片内，片外速率限制因素增多
- 采样困难
 - 模拟信号 - 电压和时钟两个维度
 - 眼图小，有效采样范围小
 - 温度变化导致走线延迟变化
 - 电源电压不稳导致的时钟抖动
 - 传输线上的负载平衡



DRAM的特点带来的设计挑战?

DRAM是动态存储器

- 需要通过刷新操作来维持存储数据的存在
- 需要DDR控制器定期发送刷新命令给DRAM
- DRAM可以进行自刷新，即定期自行刷新

Table 235 — Refresh Requirement Parameters for BG mode or 16Bank mode

Refresh Requirements		Symbol	2Gb	3Gb	4Gb	6Gb	8Gb	12Gb	16Gb	24Gb	32Gb	Units
Number of banks per channel			16									
Refresh Window (tREFW) (1x Refresh) ^{2,3}		tREFW	32									ms
Required Number of REFRESH Commands in a tREFW window		R	8192	8192	8192	8192	8192	8192	8192	8192	8192	
Average Refresh Interval (1x Refresh) ²	REFab	tREFI	3.906	3.906	3.906	3.906	3.906	3.906	3.906	3.906	3.906	us
	REFpb	tREFipb	488	488	488	488	488	488	488	488	488	ns
Refresh Cycle Time (All Banks)		tRFCab	130	180	180	210	210	280	280	380	380	ns
Refresh Cycle time (Per Bank)		tRFCpb	60	90	90	120	120	140	140	190	190	ns
Per-bank Refresh to Per-bank Refresh different bank time		tpbR2pb R	60	90	90	90	90	90	90	90	90	ns
REFpb to Activate command to different bank		tpbR2act	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	ns
Refresh Management Cycle time (All Banks)		tRFMab	-	-	-	-	-	-	-	380	380	ns
Refresh Management Cycle time (Per Bank)		tRFMpb	-	-	-	-	-	-	-	260	260	ns
NOTE 1 Refresh for each channel is independent of the other channel in a package. Power delivery in the user's system should be verified to make sure the DC operating conditions are maintained when multiple channels are refreshed simultaneously.												
NOTE 2 1x refresh rate (tREFW=32ms) is supported at all temperatures at or below 85°C Tcase. If MR4 OP[4:0] indicates a refresh rate of greater than 1x is supported, tREFW can be extended.												
NOTE 3 Refer to MR4 OP[4:0] (Table 68) for detailed Refresh Rate and its multipliers.												

初步了解DRAM Jedec Spec

概况

JEDEC Standard No. 209-5B

LPDDR5/LPDDR5X SDRAM is a high-speed synchronous SDRAM device internally configured with 1 channel containing either 16 or 8 DQ signals. The bank architecture is user-selectable and can be either eight banks (8B Mode), four banks with four bank groups (BG Mode), or sixteen banks (16B Mode) for LPDDR5 SDRAM, and either four banks with four bank groups (BG Mode), or sixteen banks (16B Mode) for LPDDR5X SDRAM. See 2.2.3 for more information.

Density can range from 2Gb to 32Gb.

Data access size depends on the bank configuration, with 16n and 32n prefetches supported. A single BL16 Read or Write access for a x16, BG mode LPDDR5 SDRAM, typically consists of a single 256-bit-wide data transfer at the internal SDRAM core, and sixteen corresponding 16-bit wide, one half-WCK-clock-cycle data transfers at the I/O pins.

LPDDR5 SDRAM devices use a command clock (CK) that operates at a reduced rate from per-byte data clock (WCK). There are seven (DDR) command/address (CA) pins that the memory controller uses to transmit command, address, bank, configuration, and training information to the SDRAM. CA signals are latched on both the rising and falling CK edges when indicated by a high signal on the single (SDR) CS pin. Most commands are 1 nCK in duration. See Table 197 for details.

The WCK:CK ratio is user-selectable as either 2:1 or 4:1. For low-power operation the WCK is generally designed to operate only when read or write data needs to be transmitted on the bus. Due to the high speeds required for WCK, the LPDDR5 SDRAM generally implements circuitry to divide the WCK frequency immediately after the WCK receiver. This leads to a requirement to synchronize WCK to CK when the WCK needs to be re-started from an idle time. See 7.2.1 for details.

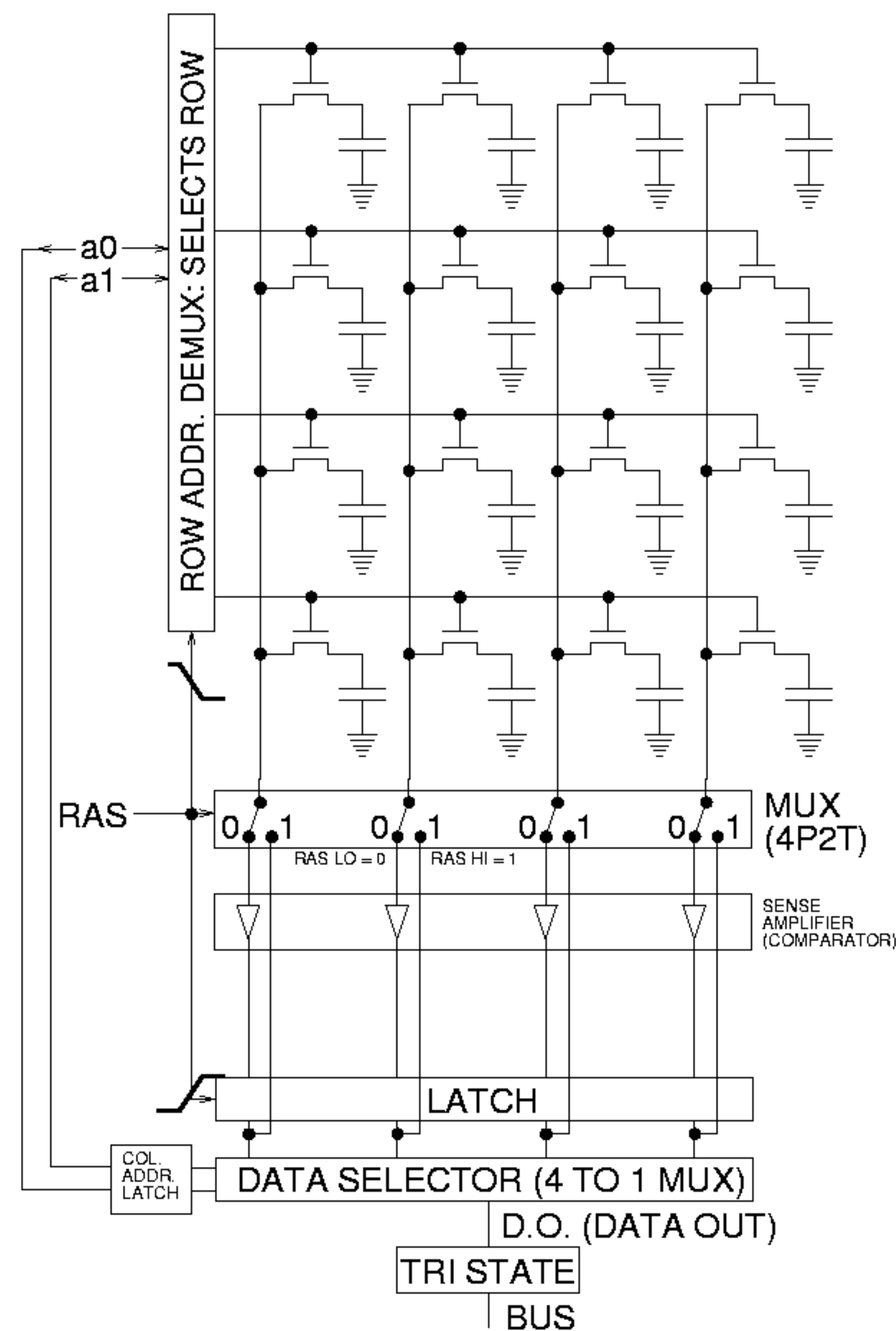
初步了解DRAM Jedec Spec

基本概念

Read and Write accesses to LPDDR5 SDRAMs are burst oriented; accesses start at a selected location and continue for a programmed number of locations in a programmed sequence. Accesses begin with the registration of an Activate command. Prior to issuing a Read, Write, or Mask Write command, WCK must be synchronized to CK by issuing a CAS command with (WS_RD, WS_WR or WS_FS) followed by a Read, Write, or Mask Write command.

The address and BA bits registered coincident with the Activate command are used to select the row and the bank to be accessed. The address bits registered coincident with the Read, Write or Mask Write command are used to select the bank and the starting column location for the burst access.

Prior to normal operation, the LPDDR5 SDRAM is required to be initialized. The Power-Up, Initialization, and Power-off Procedure section provides detailed information covering device initialization.



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端口信号

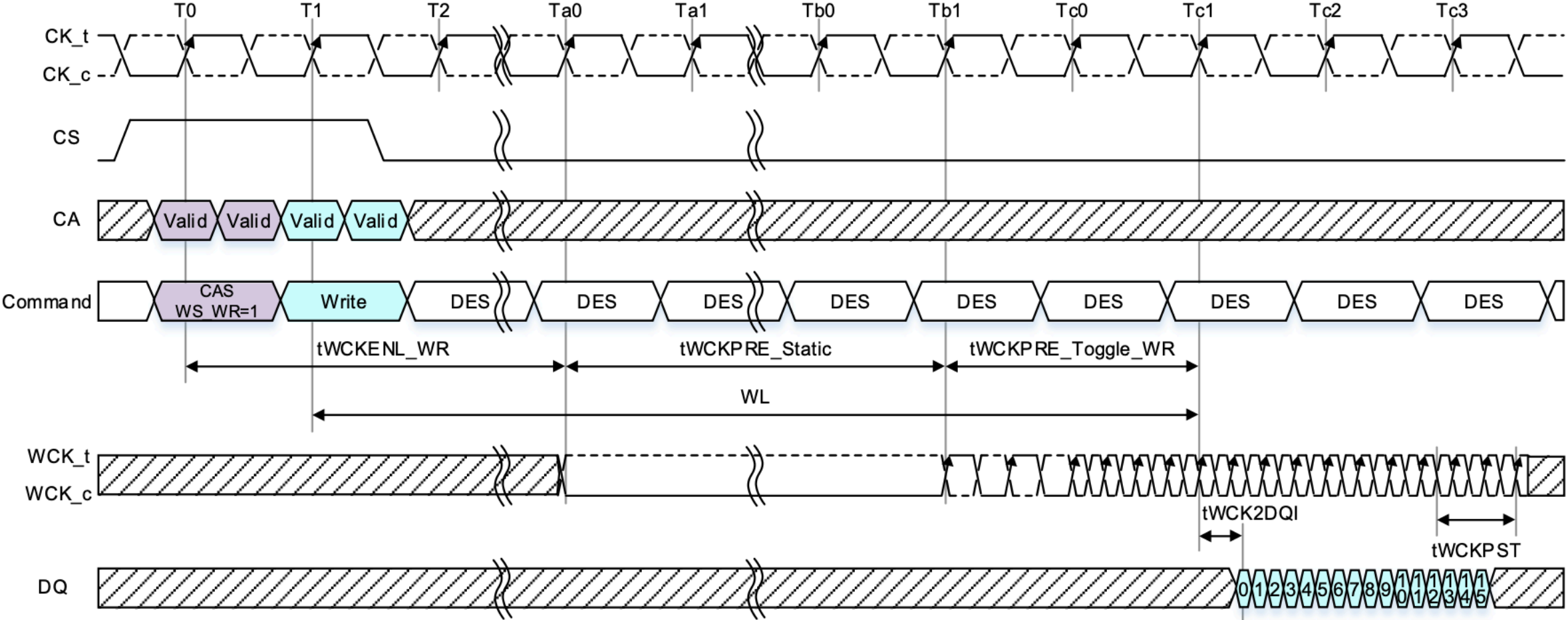


Table 1 — Pad Definition and Description

Symbol	Type	Description	Note
CK_t, CK_c	Input	Clock: CK_t and CK_c are differential clock inputs. All Double Data Rate (DDR) Command/Address inputs are sampled on both crossing points of CK_t and CK_c. The first crossing point is the rising(falling) edge of CK_t (CK_c) and second crossing point is falling(rising) edge of CK_t (CK_c). Single Data Rate (SDR) inputs, CS is sampled on the crossing point that is the rising(falling) edge of CK_t (CK_c).	
CS	Input	Chip Select: CS is part of the command code and is sampled on the rising(falling) edge of CK_t (CK_c) unless the device is in power-down or Deep Sleep mode where it becomes an asynchronous signal.	
CA[6:0]	Input	Command/Address Inputs: CA signals provide the Command and Address input according to the Command Truth Table	
DQ[15:0]	I/O	Data Input/Output: Bi-direction data bus.	
WCK[1:0]_t WCK[1:0]_c	Input	Data Clocks: WCK_t and WCK_c are differential clocks used for WRITE data capture and READ data output.	
RDQS [1:0]_t , RDQS[1:0]_c	RDQS_t :I/O, RDQS_c :Output	Read Data Strobe: RDQS_t and RDQS_c are the differential output clock signals used to strobe data during a READ operation. And RDQS_t is also used as a Parity pin at Write with Link Protection enabled.	1
DMI[1:0]	I/O	Data Mask Inversion: DMI achieves multiple function such as Data Mask (DM), Data Bus Inversion (DBI), and Parity at read with ECC operation by setting the Mode Register and DMI is a bi-directional signal, and each byte of data has a DMI signal.	1
ZQ	Reference	ZQ: ZQ is used to calibrate the output drive strength and the termination resistance as calibration reference. There is one ZQ pad per die. The ZQ pin shall be connected to VDDQ through a 240Ω ±1% resistor.	
VDDQ, VDD1, VDD2H, VDD2L	Supply	Power Supplies: Isolated on the die for improved noise immunity.	
VSS	GND	Ground Reference: Power supply ground reference	
RESET_n	Input	RESET: When asserted LOW, the RESET_n signal resets the die. RESET_n is an asynchronous signal.	
NOTE 1 See Table 2 and Table 3, Per-Byte Signal List/Description, for multi-function definition for these pins.			

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内部结构 - Bank Architecture

- Bank Group用于弥补存储单元速度较低的缺陷
- 可以通过增加Burst Length来弥补这个缺陷
- 但是增加Burst Length会降低灵活性，即最小的数据访问单元会增加

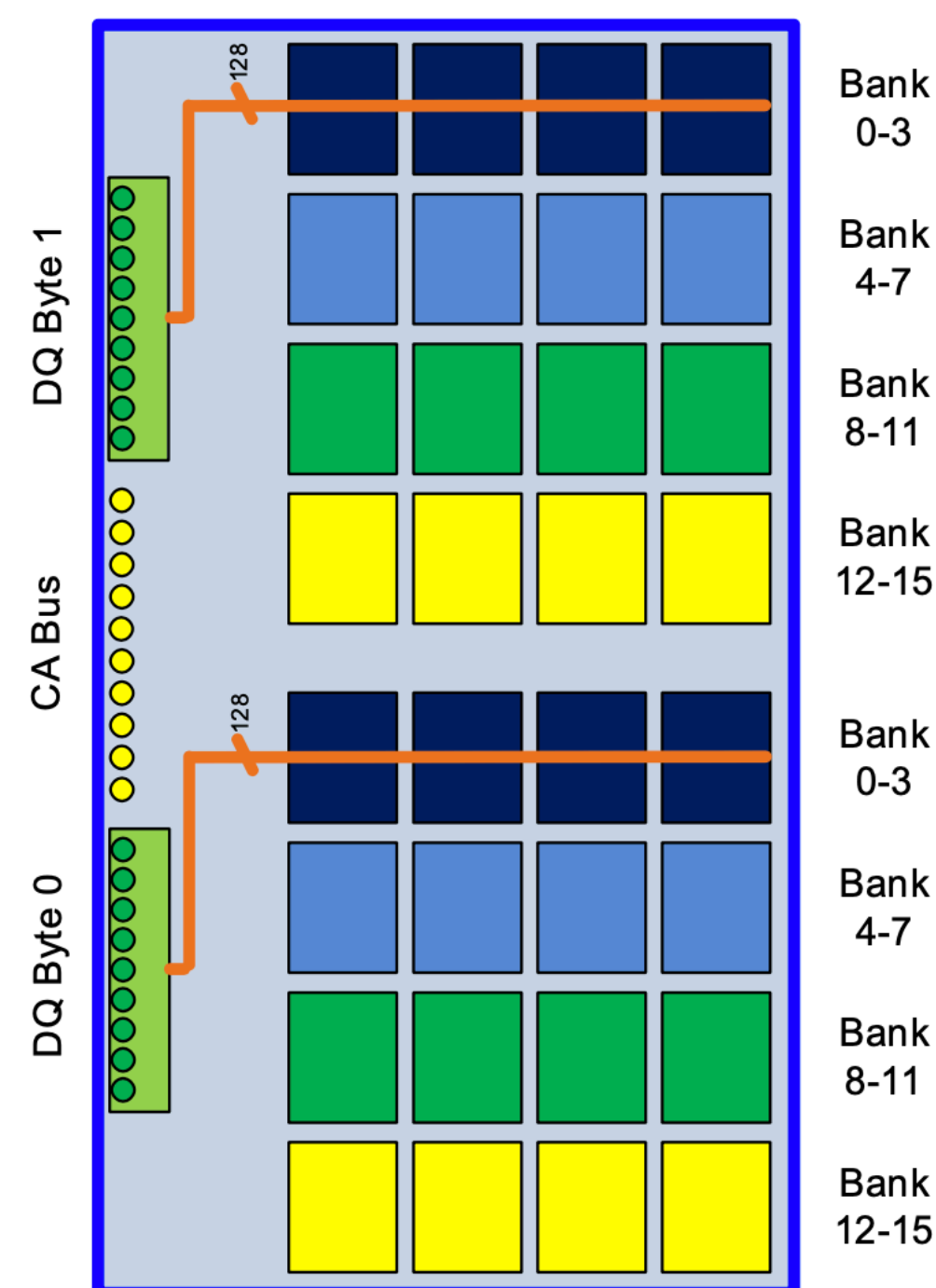


Figure 3 — 16B Mode Configuration Example (One Channel Shown)

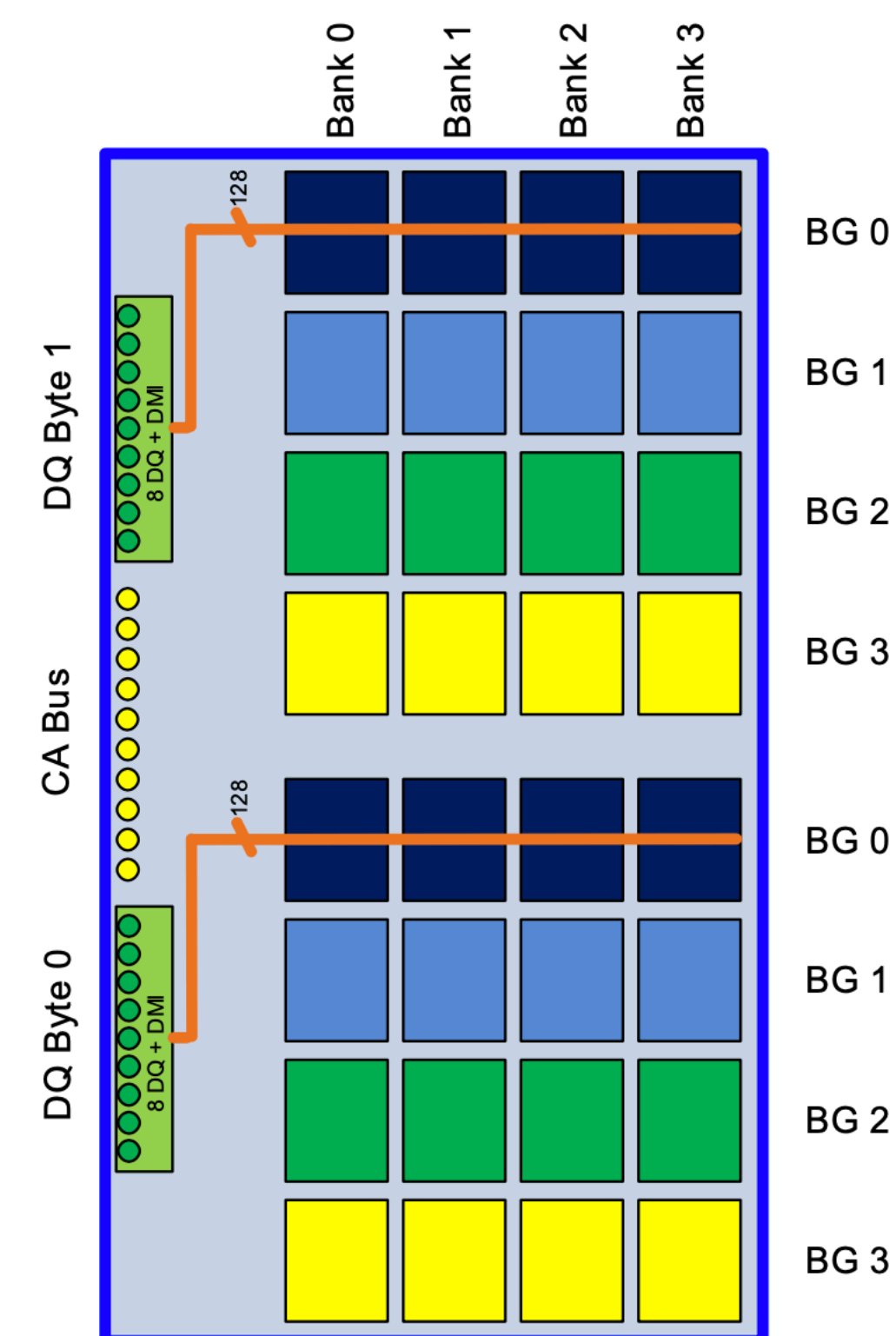


Figure 1 — BG Mode Configuration Example (One Channel Shown)

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命令

- 真值表

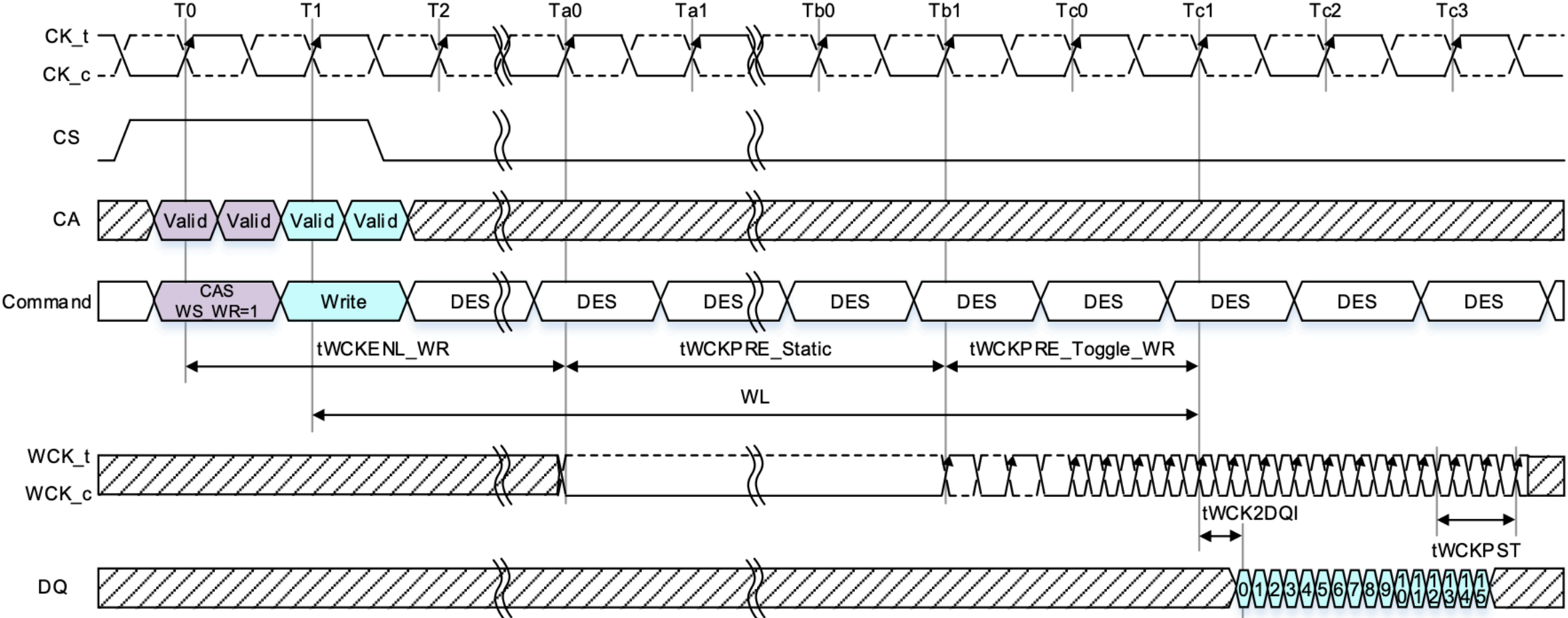


Table 197 — Command Truth Table

SDRAM COMMAND	BK ORG (BG, 16B, 8B)	SDR CMD PIN	DDR COMMAND PINS							CK_t edge	Notes
		CS	CA0	CA1	CA2	CA3	CA4	CA5	CA6		
DESELECT (DES)	Any	L	X	X	X	X	X	X	X	R1	1, 2
		X	X	X	X	X	X	X	X	F1	
NO OPERATION (NOP)	Any	H	L	L	L	L	L	L	L	R1	1, 2
		X	X	X	X	X	X	X	X	F1	
POWER DOWN ENTRY (PDE)	Any	H	L	L	L	L	L	L	H	R1	1, 2, 12
		L	X	X	X	X	X	X	X	R2	
ACTIVATE-1 (ACT-1)	Any	H	H	H	H	R14	R15	R16	R17	R1	1, 2, 3, 4
	BG	X	BA0	BA1	BG0	BG1	R11	R12	R13	F1	
	16B	X	BA0	BA1	BA2	BA3				F1	
ACTIVATE-2 (ACT-2)	Any	H	H	H	L	R7	R8	R9	R10	R1	1, 2, 4
		X	R0	R1	R2	R3	R4	R5	R6	F1	
PRECHARGE (PRE) (Per Bank, All Banks)	Any	H	L	L	L	H	H	H	H	R1	1, 2, 3, 5
	BG	X	BA0	BA1	BG0	BG1	V	V	AB	F1	
	16B	X	BA0	BA1	BA2	BA3				F1	
REFRESH (REF) (Per Bank, All Banks)	Any	H	L	L	L	H	H	H	L	R1	1, 2, 3, 5, 14
	BG	X	BA0	BA1	BG0	BG1	SB0	V	AB	F1	
	16B	X	BA0	BA1	BA2	BA3				F1	
MASK WRITE (MWR)	Any	H	L	H	L	C0	C3	C4	C5	R1	1, 2, 3, 6, 7
	BG	X	BA0	BA1	BG0	BG1	C1	C2	AP	F1	
	16B	X	BA0	BA1	BA2	BA3				F1	
WRITE (WR16 or WR)	Any	H	L	H	H	C0	C3	C4	C5	R1	1, 2, 3, 6, 7, 8
	BG	X	BA0	BA1	BG0	BG1	C1	C2	AP	F1	
	16B	X	BA0	BA1	BA2	BA3				F1	
WRITE32 (WR32)	BG/16B	H	L	L	H	L	C3	C4	C5	R1	1, 2, 3, 6, 7, 8
	BG	X	BA0	BA1	BG0	BG1	C1	C2	AP	F1	
	16B	X	BA0	BA1	BA2	BA3				F1	
READ (RD16 or RD)	Any	H	H	L	L	C0	C3	C4	C5	R1	1, 2, 3, 6, 7, 10, 11
	BG	X	BA0	BA1	BG0	BG1	C1	C2	AP	F1	
	16B	X	BA0	BA1	BA2	BA3				F1	
READ32 (RD32)	BG/16B	H	H	L	H	C0	C3	C4	C5	R1	1, 2, 3, 6, 7, 10, 11
	BG	X	BA0	BA1	BG0	BG1	C1	C2	AP	F1	
	16B	X	BA0	BA1	BA2	BA3				F1	

命令

- 真值表

[illegible]

SDRAM COMMAND	BK ORG (BG, 16B, 8B)	SDR CMD PIN	DDR COMMAND PINS							CK_t edge	Notes
		CS	CA0	CA1	CA2	CA3	CA4	CA5	CA6		
CAS	Any	H	L	L	H	H	WS_WR	WS_RD	WS_FS	R1	1, 2, 7, 13
		X	DC0	DC1	DC2	DC3	WRX	WXSA	WXSB/B3	F1	
MULTI PURPOSE COMMAND (MPC)	Any	H	L	L	L	L	H	H	OP7	R1	1, 2
		X	OP0	OP1	OP2	OP3	OP4	OP5	OP6	F1	
SELF REFRESH ENTRY (SRE)	Any	H	L	L	L	H	L	H	H	R1	1, 2, 9
		X	V	V	V	V	V	DSM	PD	F1	
SELF REFRESH EXIT (SRX)	Any	H	L	L	L	H	L	H	L	R1	1, 2
		X	V	V	V	V	V	V	V	F1	
MODE REGISTER WRITE-1 (MRW-1)	Any	H	L	L	L	H	H	L	H	R1	1, 2
		X	MA0	MA1	MA2	MA3	MA4	MA5	MA6	F1	
MODE REGISTER WRITE-2 (MRW-2)	Any	H	L	L	L	H	L	L	OP7	R1	1, 2
		X	OP0	OP1	OP2	OP3	OP4	OP5	OP6	F1	
MODE REGISTER READ (MRR)	Any	H	L	L	L	H	H	L	L	R1	1, 2
		X	MA0	MA1	MA2	MA3	MA4	MA5	MA6	F1	
WRITE FIFO (WFF)	Any	H	L	L	L	L	L	H	H	R1	1, 2
		X	L	L	L	L	L	L	L	F1	
READ FIFO (RFF)	Any	H	L	L	L	L	L	H	L	R1	1, 2
		X	L	L	L	L	L	L	L	F1	
READ DQ CALIBRATION (RDC)	Any	H	L	L	L	L	H	L	H	R1	1, 2
		X	L	L	L	L	L	L	L	F1	
NOTE 1 LPDDR5 commands are one clock cycles long and defined by the states of CS at the rising edge (R1) of clock and CA[6:0] at the rising edge (R1) and the falling edge (F1) of clock. Note that some operations such as ACTIVATE, MODE REGISTER WRITE or WCK2CK SYNC (WS_RD, WS_WR, WS_FS) require two commands to initiate. NOTE 2 "V" means "H" or "L" (a defined logic level). "X" means don't care in which case CK_t, CK_c and CA[6:0] can be floated. NOTE 3 Bank Group Address BG[1:0] and Bank Address BA[1:0] determine which bank is to be operated upon. NOTE 4 An ACTIVATE-1 command must be followed by an ACTIVATE-2 command. Only CAS, WRITE, MASK WRITE, READ, PRECHARGE (to a different bank) commands can be issued between ACTIVATE-1 and ACTIVATE-2 commands. The maximum timing gap between ACTIVATE-1 and ACTIVATE-2 commands are 8 clock cycles. An ACTIVATE-1 command must be issued first before issuing an ACTIVATE-2 command. Once an ACTIVATE-1 command is issued, an ACTIVATE-2 command must be issued before issuing another ACTIVATE-1 command. NOTE 5 AB "HIGH" during PRECHARGE or REFRESH command indicates that command must be applied to all banks and bank address is a don't care.											

下节预告

第三节 - DRAM Jedec Spec从用户的角度看读写

- Activate
- WCK2CK Sync
- Read / Write
- 从用户的角度看如何读取一个数据回来
- 从用户的角度看如何写一个数据进去

谢谢